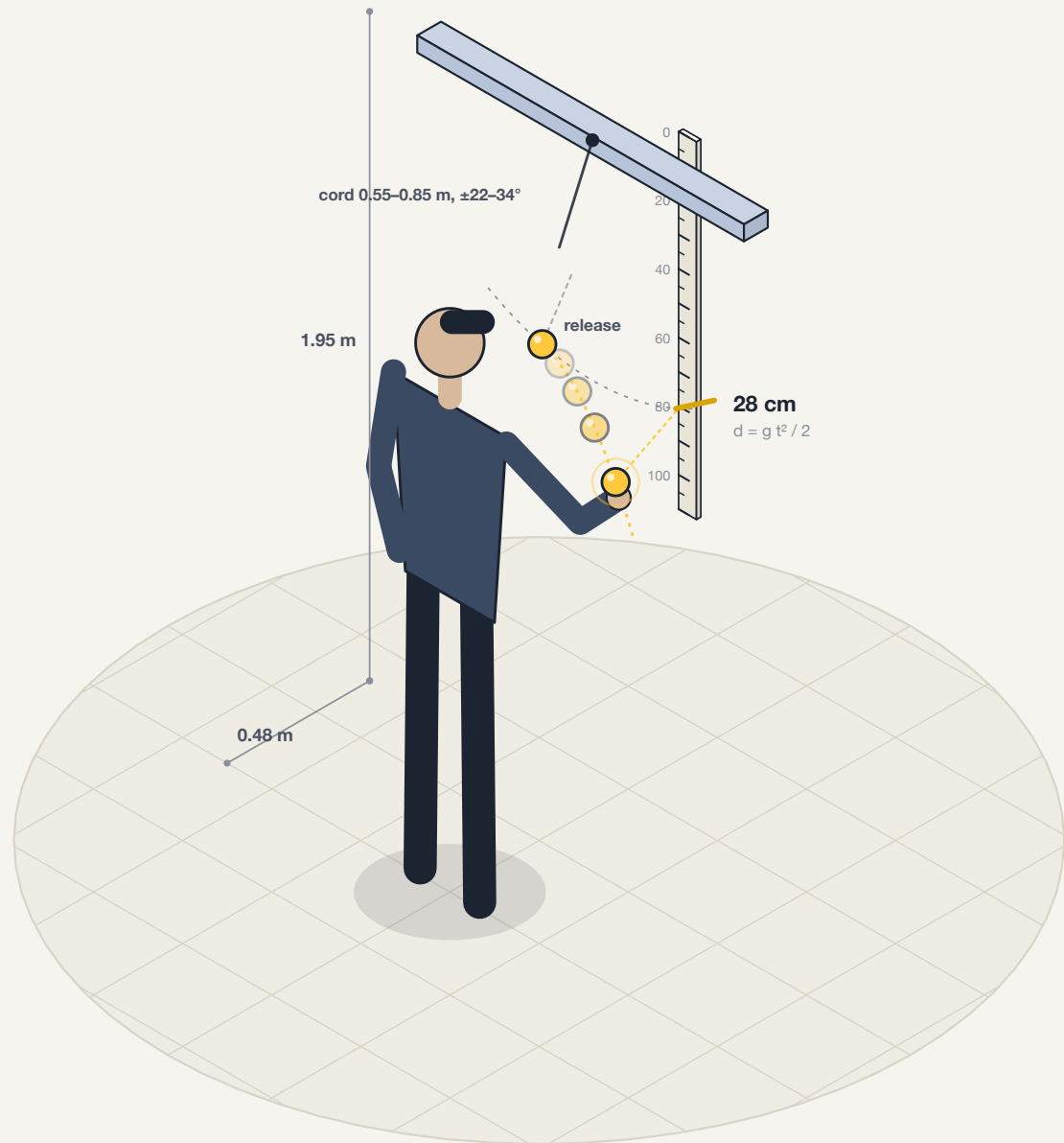


SPATIAL HUMAN ASSESSMENT

Spatial Age

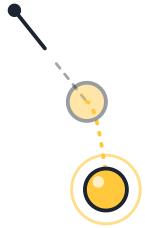
Five reaction minigames for Apple Vision Pro. Life scale, fully immersive, scored on device into one functional age.

ScoreKit 0.2.0 · norms prior-0.2 · schema 0.2.0



Five games, one score

Each game runs a short unscored practice round, then its scored trials. About six minutes end to end.



● Pendulum

Catch the weight when the cord lets go.

Catch latency and drop distance

12 SCORED · 3 PRACTICE

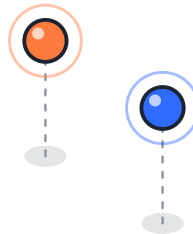


● Spark

Touch each light as it appears.

Reaction, movement and reach kinematics

20 SCORED · 5 PRACTICE

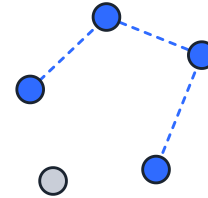


● Gate

Touch blue. Leave orange.

Decision time and inhibition

30 SCORED · 5 PRACTICE

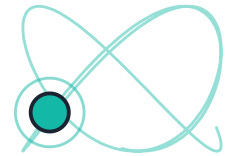


● Constellation

Watch the stars light, then touch them in order.

Spatial working memory span

14 SCORED · 1 PRACTICE



● Orbit

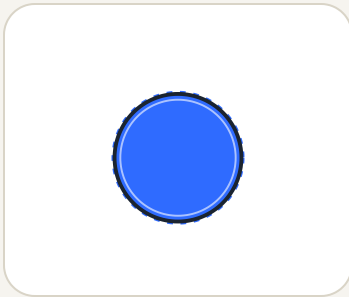
Keep your fingertip inside the moving light.

Visuomotor tracking error and lag

3 SCORED · 1 PRACTICE

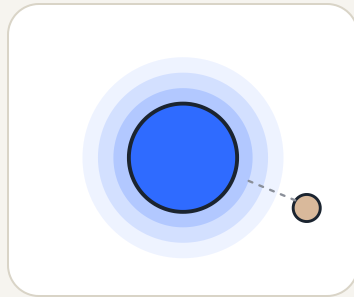
Five primitives

Every game is built from the same five motions. Feedback confirms contact. It never reveals timing or score.



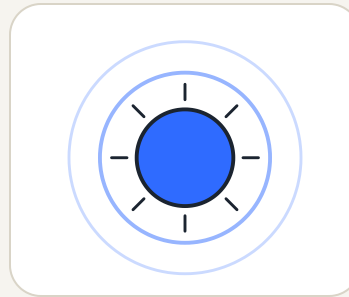
Breathe

Live targets pulse 1.00 to 1.03 at 0.5 Hz.



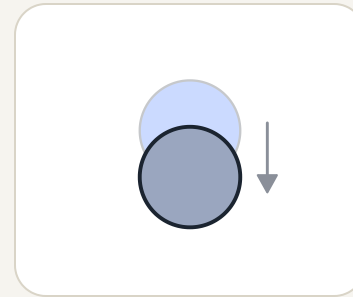
Proximity glow

Emissive ramps as the fingertip closes from 30 cm.



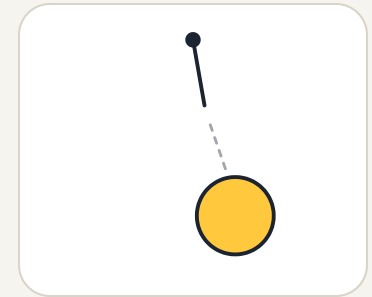
Contact pop

90 ms scale 1 → 1.25 → 0. Ring grows to 3× radius over 240 ms. Tick pitch follows speed.



Miss sink

250 ms desaturate, drop 3 cm, fade.



Release snap

Cord flicks and fades in 120 ms with a low tick.

SPACE

Full immersion. Ink sky dome and a 6 m floor grid disc. Nothing else in view.

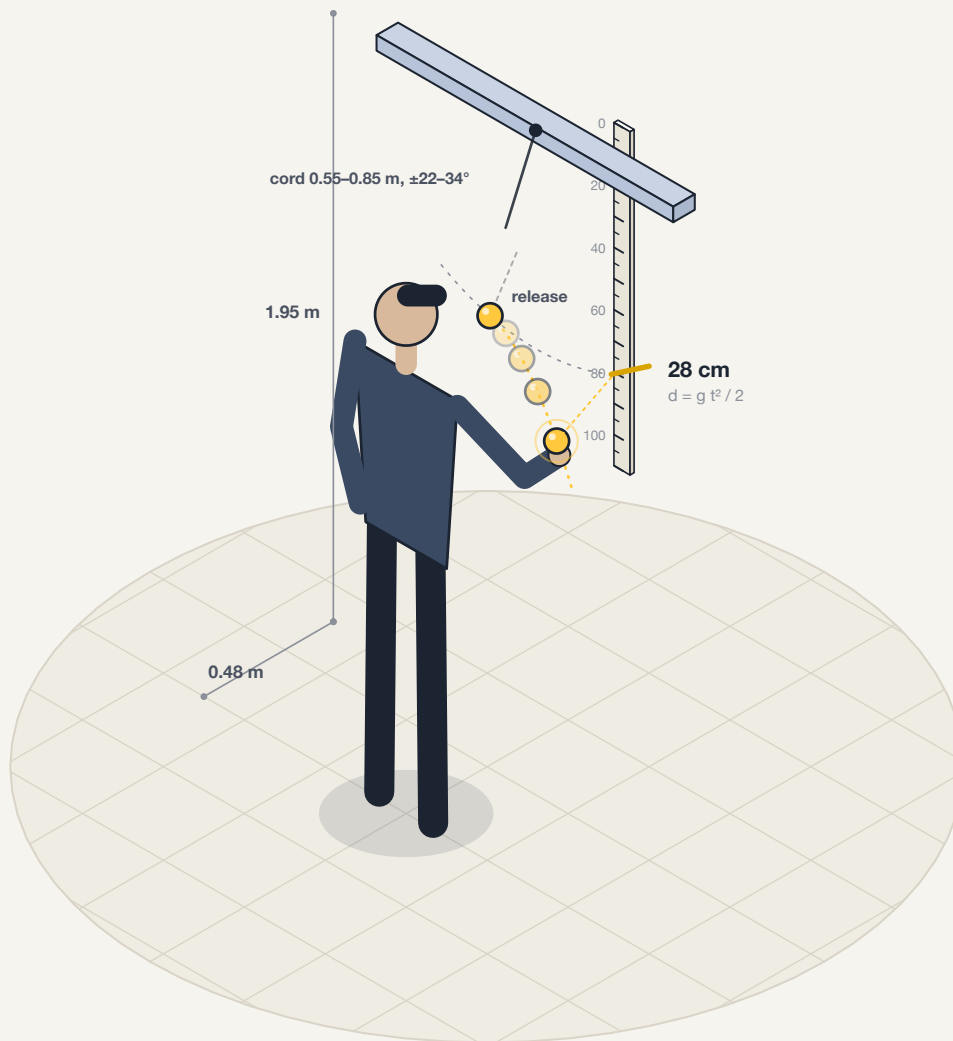
SCALE

Every object at true size and distance, in meters from the participant.

SCORE

Hidden during play. Shown once, after the last game.

Pendulum



Catch the weight when the cord lets go.

MECHANIC

A gold bob, 8 cm, swings on a 0.55 to 0.85 m cord from a pivot 0.4 m above eye level (1.95 m standing), 0.48 m ahead.

Swing amplitude 22 to 34°. The cord releases at a random time, 1.5 to 4 s into the swing.

The bob falls ballistically at 9.81 m/s² from its release position and velocity. Catch when the thumb-index midpoint is within 7 cm of the bob with the pinch closed under 4.5 cm.

A life-size cm ruler stands behind the swing plane. A gold tick marks the catch height.

MICRO-INTERACTIONS

Release snap on the cord. Contact pop on catch, with the bob frozen in the hand. Miss sink when it passes the floor.

CAPTURED PER TRIAL

length_m, amplitude_deg, release_t, release_angle_deg, release_position, release_velocity, catch_t, catch_position, hand, outcome, aperture_release_m, aperture_catch_m, tracking_gap_ms, trace

METRICS

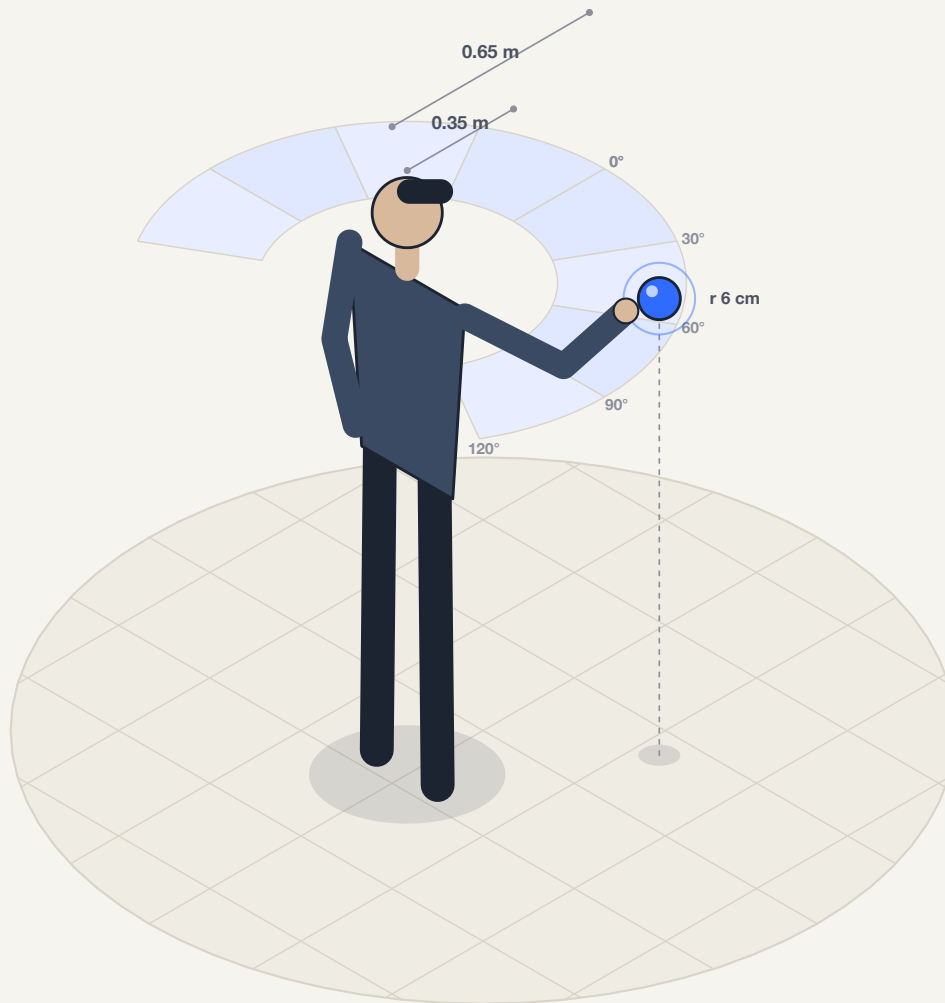
Catch latency catch_t – release_t, median of catches

Drop distance vertical fall at catch, cm. The ruler drop test, $d = g t^2 / 2$

Catch rate catches / attempts

Anticipation grasp under 100 ms after release, excluded

Spark



Touch each light as it appears.

MECHANIC

One blue sphere, radius 6 cm, 0.35 to 0.65 m from the shoulder.

Eccentricity 0 to 120° from head forward, balanced over 4 bins.

Random foreperiod 0.8 to 2.0 s. Trial ends on fingertip contact or a 3 s timeout.

MICRO-INTERACTIONS

Breathe while live. Proximity glow on approach. Contact pop with a tick pitched by speed. Miss sink on timeout.

CAPTURED PER TRIAL

kind, spawn_t, move_t, contact_t, position, eccentricity_deg, outcome, hand, endpoint_error_m, tracking_gap_ms, trace (fingertip path, 90 Hz)

METRICS

Reaction time trace onset – spawn

Movement time contact – onset

RT tail ex-Gaussian tau

RT variability SD / mean, trimmed

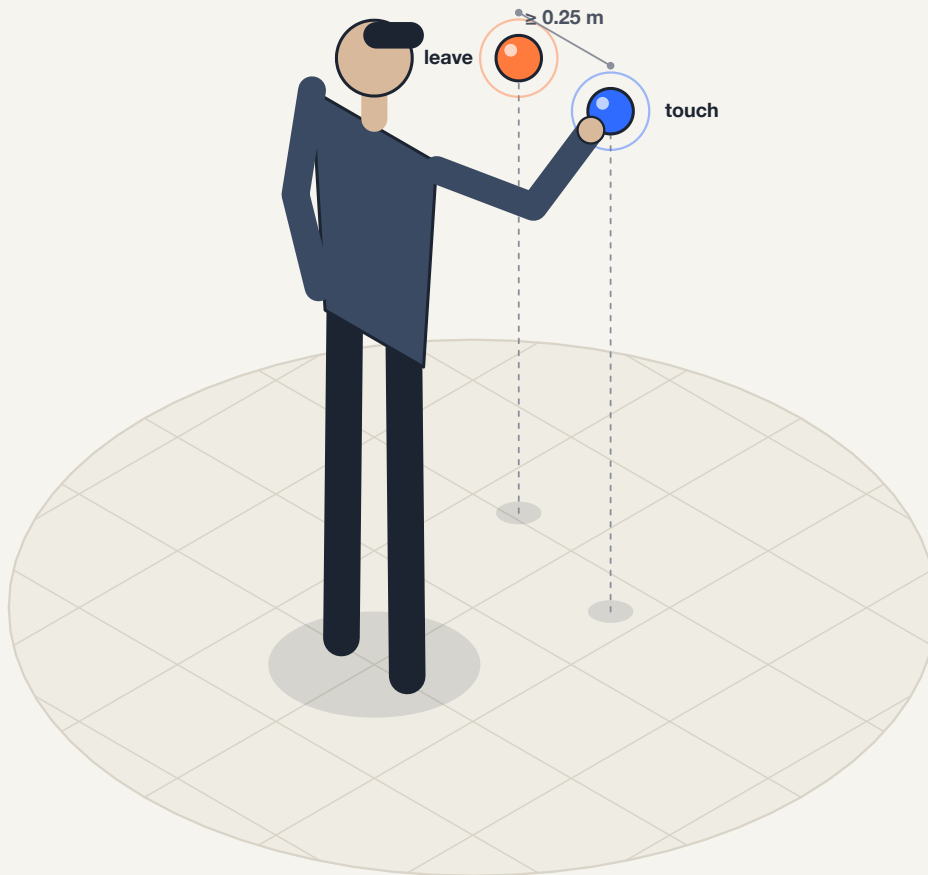
Eccentricity cost slope of reach time per 90°

Peak speed fingertip, m/s

Path efficiency straight line / path length

Smoothness log dimensionless jerk

Gate



Touch blue. Leave orange.

MECHANIC

Go trials, 70%: blue and orange appear together, at least 25 cm apart. Touch blue.

No-go trials, 30%: orange alone. Hold still.

Response window 2.0 s. Placement and timing as Spark.

MICRO-INTERACTIONS

Orange never glows on approach. A correct hold dissolves orange quietly. Touching it plays miss sink.

CAPTURED PER TRIAL

As Spark. kind is go or nogo. outcome is hit, miss, false_alarm or correct_reject.

METRICS

Choice RT median, go hits

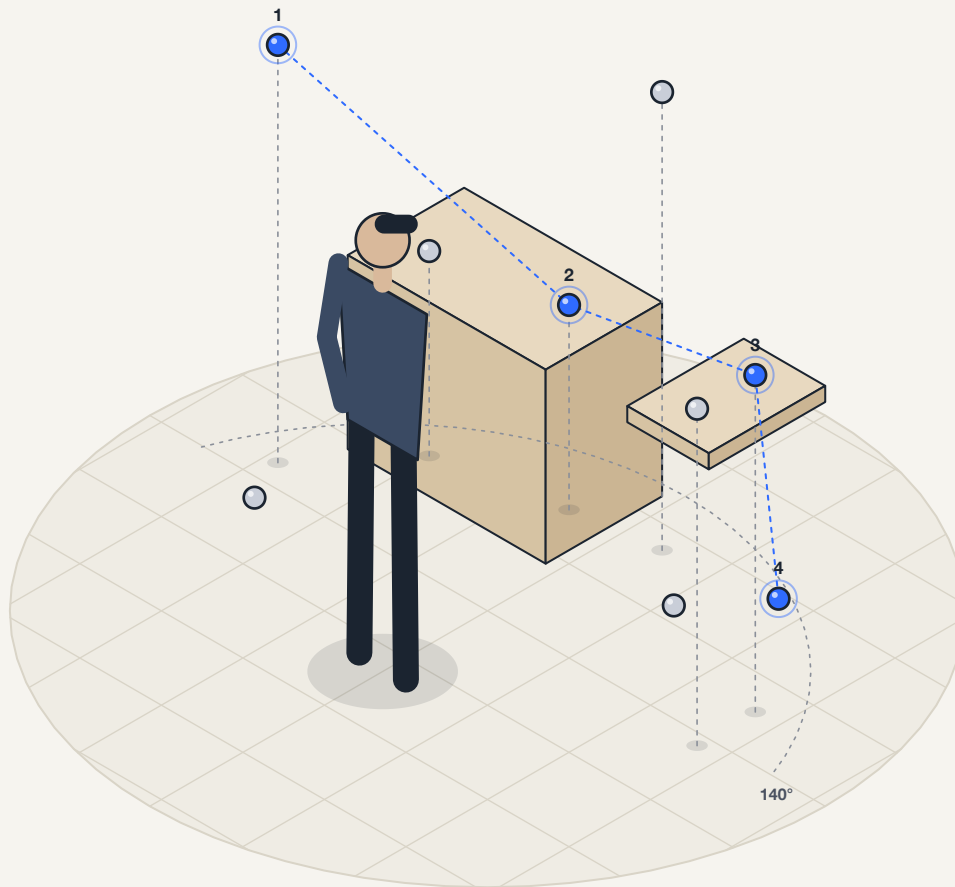
Decision time choice RT – Spark RT

Commission false alarms / trials

Omission misses / go trials

Sensitivity d' $z(\text{hit}) - z(\text{false alarm})$, log-linear

Constellation



Watch the stars light, then touch them in order.

MECHANIC

9 star nodes, 8 cm, anchored on room surfaces across 140° of azimuth. Some need a head turn.

A sequence lights, 0.8 s on, 0.2 s gap. Touch it back in order.

Staircase from span 2: two correct steps up, two fails stop. Cap at span 9.

MICRO-INTERACTIONS

Nodes breathe while waiting. Each touch flashes white and pops. A wrong touch sinks the whole set.

CAPTURED PER TRIAL

span, sequence, response, correct, start_t, end_t, tap_t, seed

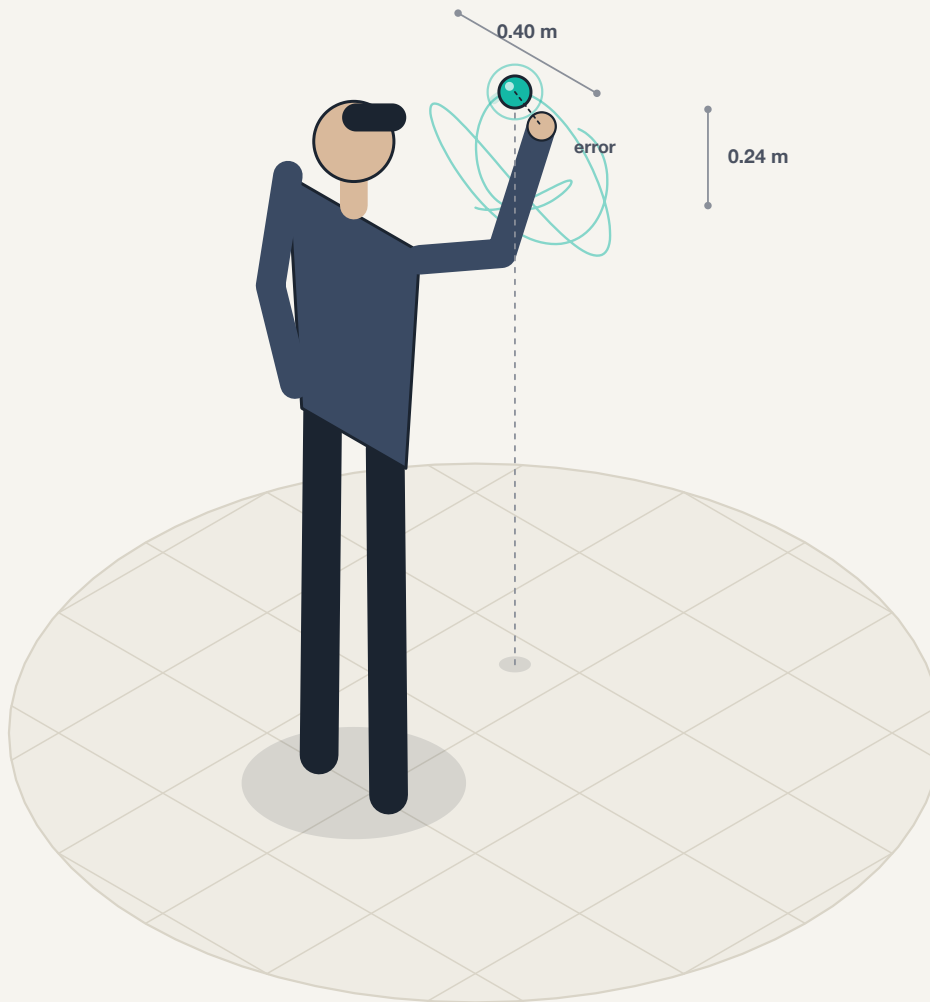
METRICS

Corsi span longest span with a correct sequence

Total score span × correct sequences (Kessels)

Tap interval median gap between response touches

Orbit



Keep your fingertip inside the moving light.

MECHANIC

A teal orb, 4 cm, follows a 3D Lissajous path centered 1.32 m high, 0.45 m ahead.

Amplitude 0.20, 0.12, 0.08 m at 0.21, 0.29, 0.13 Hz.

3 trials of 12 s. The first second is acquisition and is not scored. A faint 0.3 s trail leads the orb.

MICRO-INTERACTIONS

Proximity glow holds while the fingertip is inside. It dims the moment the finger leaves.

CAPTURED PER TRIAL

start_t, duration_s, path (center, amplitude, frequency_hz, phase), t, target, finger (null while untracked), hand, tracking_gap_ms

METRICS

Tracking error RMS fingertip to orb, cm

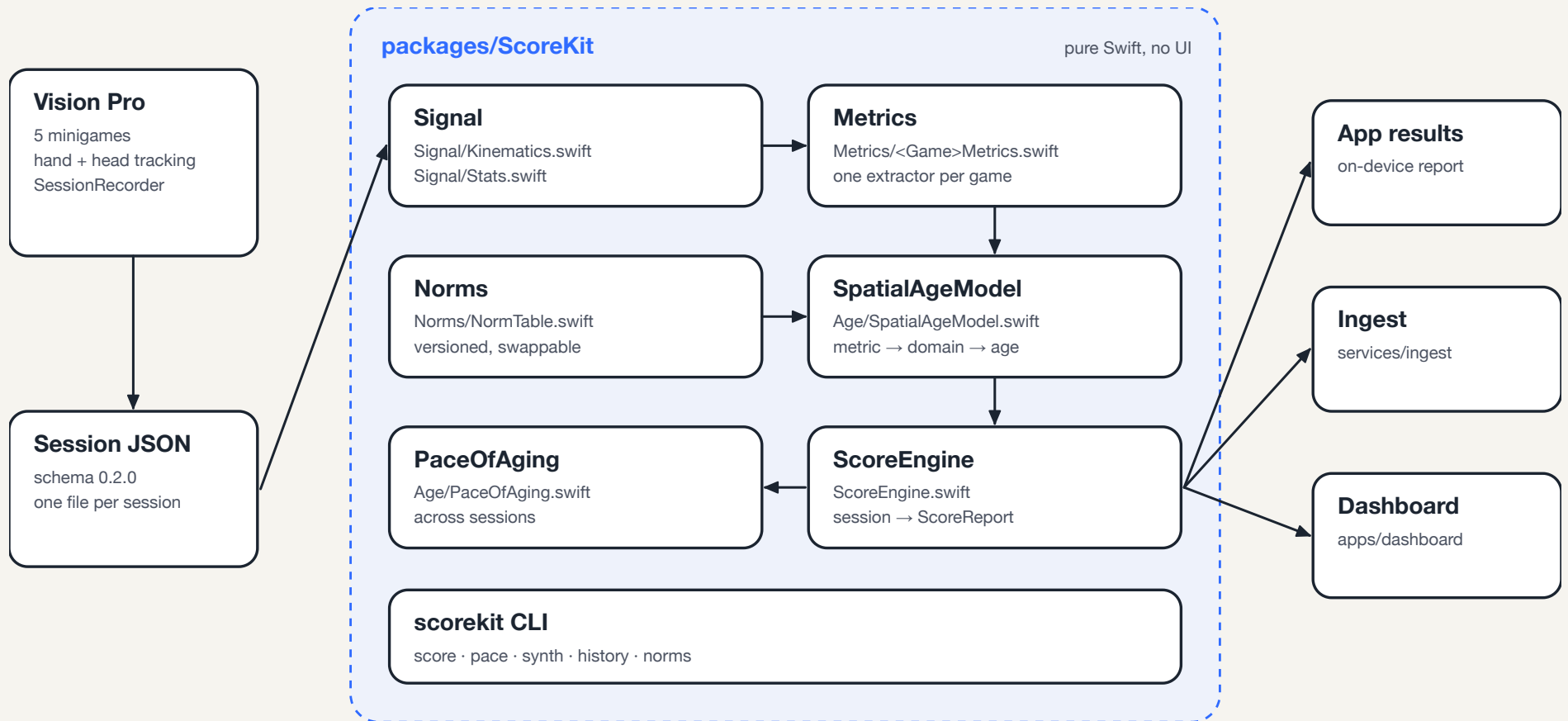
Lag path shift that best fits the fingertip, ms

Time on target share of samples within 4 cm

Velocity gain fingertip velocity projected on orb velocity

ScoreKit

Scoring lives in one Swift package with no UI or ARKit dependency. The app, the CLI and the tests link the same code.



From fingertip to age

Movement onset

Trace resampled to 90 Hz, Gaussian smoothed (σ 2 samples). Onset fires when speed stays above 0.15 m/s for 50 ms, then walks back to where speed last fell under 5% of peak.

$$RT = \text{onset} - \text{spawn_t}, MT = \text{contact_t} - \text{onset}$$

RT distribution

Ex-Gaussian by method of moments on untrimmed RTs (0.10 to 1.50 s). Tau is the slow tail. Medians use 3-MAD trimming.

$$\tau = s \cdot (\gamma/2)^{1/3}, \mu = \text{mean} - \tau$$

Metric age

Each norm is a curve $m(a) = m_{25} + \beta(a-25) + \gamma \cdot \max(0, a-50)^2$. A value inverts to an age by bisection over 18 to 95. Its SD combines measurement and model error.

$$sd = \sqrt{(\text{sem} / |m'(a)|)^2 + \tau^2}$$

Spatial Age

Evidence combines with a prior centered on chronological age, SD 9 years. Thin evidence stays near the calendar. Reported with an 80% interval.

$$\text{age} = (e \cdot w_e + c \cdot w_p) / (w_e + w_p)$$

Smoothness

Log dimensionless jerk on the velocity vector between onset and contact (Balasubramanian 2015). Closer to zero is smoother.

$$LDLJ = -\ln(T^3 / v_{\text{peak}}^2 \cdot \int |d^2v/dt^2|^2 dt)$$

Inhibition

Signal detection on Gate with the log-linear correction (Hautus 1995).

$$d' = \Phi^{-1}((H+.5)/(G+1)) - \Phi^{-1}((F+.5)/(N+1))$$

Fusion

Inverse-variance weights into 5 domains, then across domains. Correlated inputs inflate the variance by the design effect, ρ 0.5 within a domain and 0.3 between.

$$\text{var} = (1 + (k-1)\rho) / \sum w_i$$

Pace of aging

Weighted least squares of Spatial Age on time, at least 3 sessions over 28 days, shrunk toward 1.0 with prior SD 0.5.

$$\text{pace} = (1 \cdot w_0 + b \cdot w_b) / (w_0 + w_b)$$

Everything we capture

One JSON file per session: participant code, age, sex, handedness, device, then one block per game. Times in seconds from session start, positions in meters, y up. ? marks optional fields.

Spark and Gate · reaction trial

index	trial number in block
kind	go or nogo
hand ?	left, right or null
spawn_t	target visible
move_t	device movement onset
contact_t	fingertip entered radius
position ?	target center
eccentricity_deg	head forward to target at spawn
outcome	trial result
tracking_gap_ms ?	longest hand gap
endpoint_error_m ?	fingertip to center at contact
trace ?	fingertip path {t, p} at 90 Hz

Constellation · Corsi trial

index	trial number in block
span	sequence length
sequence	node ids shown
response	node ids touched
correct	response equals sequence
start_t ?	trial start
end_t ?	trial end
tap_t ?	time of each touch

Pendulum trial

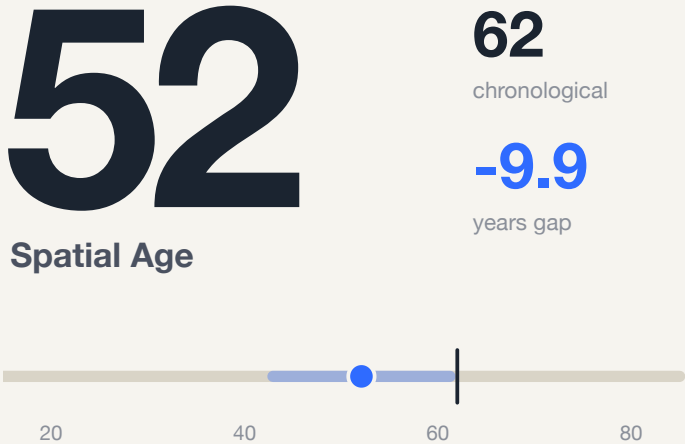
index	trial number in block
length_m	cord length
amplitude_deg	swing amplitude
release_t	cord releases
release_angle_deg	cord angle at release
release_position	bob at release
release_velocity	bob velocity at release
catch_t ?	grasp closes on bob
catch_position ?	bob at catch
hand ?	left, right or null
outcome	trial result
tracking_gap_ms	longest hand gap
aperture_release_m ?	thumb-index gap at release
aperture_catch_m ?	thumb-index gap at catch
trace ?	fingertip path {t, p} at 90 Hz

Orbit · pursuit trial

index	trial number in block
start_t	trial start
duration_s	trial length
path	Lissajous parameters
t	sample times
target	orb positions
finger	fingertip, null when untracked
hand ?	left, right or null
tracking_gap_ms	longest hand gap

One session, one number

PARTICIPANT SYN027



80% interval 43 to 61. Evidence alone 35.

GAMES

Pendulum

54

score

Drop distance 20.2 cm

Spark

45

score

Reaction time 314 ms

Gate

41

score

Decision time 123 ms

Constellation

49

score

Corsi span 6

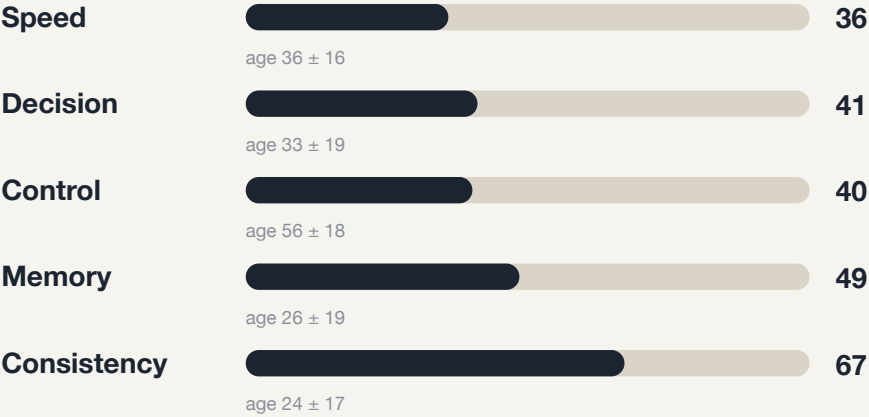
Orbit

24

score

Tracking error 5.1 cm

DOMAINS



Does it track age



Pace of aging

One synthetic participant, 52 at the start, whose latent age advances 0.8 years per calendar year. Session noise dwarfs 8 months of change, so the shrunk pace stays near 1.0. A pace claim needs a longer window or more sessions.

1.01×

pace of aging

± 0.50

posterior SD

9

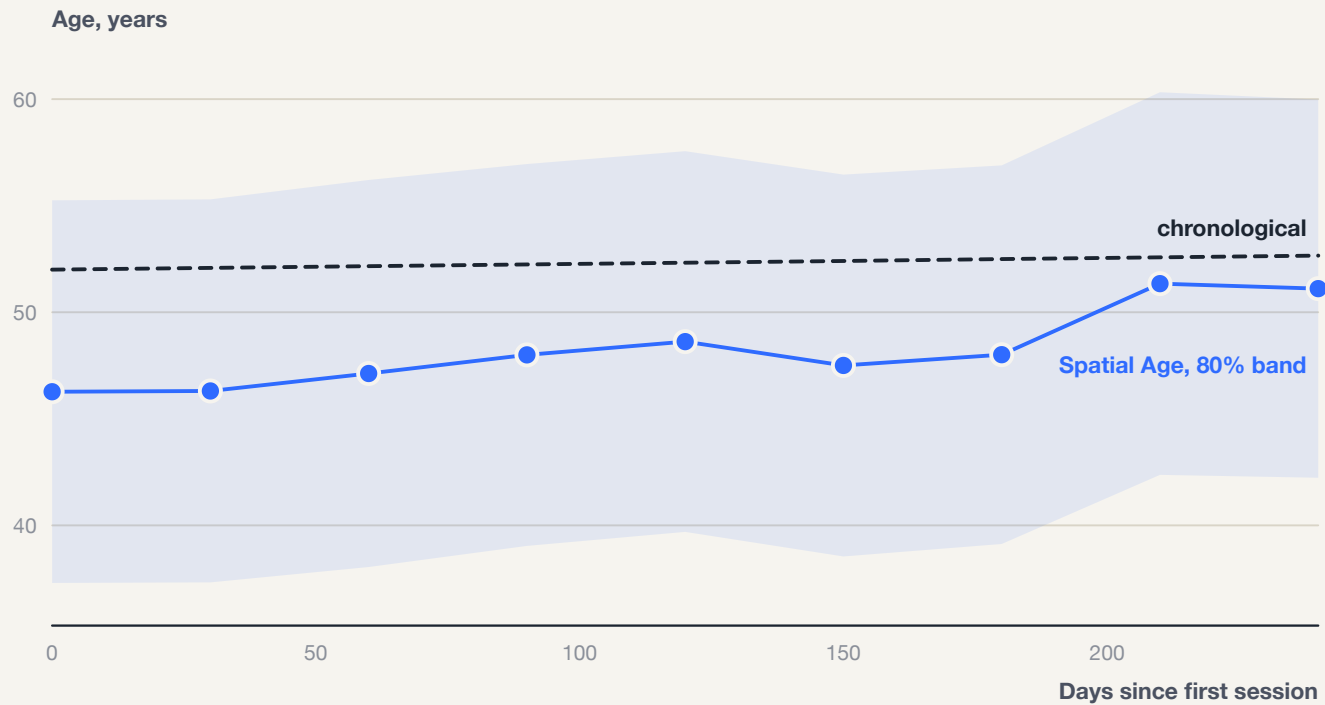
sessions

240

days

7.2

raw slope, years per year



Priors and limits

METRIC	DOMAIN	MEAN 25	SD 25	PER YEAR	MODEL SD, Y	SHAPE SOURCE
catch_latency	speed	0.19	0.025	+0.0009	14	Ruler drop RT shape: Eckner et al. 2010; age slope after Der and Deary 2006
catch_drop_cm	display	17.7	4.5	+0.17	–	Display only. Derived from catch latency, $d = g t^2 / 2$
catch_rate	control	0.95	0.05	-0.0015	26	Prior, no published reference for this task
reach_rt	speed	0.3	0.04	+0.0015	13	Simple RT age curve: Der and Deary 2006, Psychol Aging 21(1)
reach_mt	control	0.38	0.06	+0.0025	16	Reach movement slowing: Ketcham and Stelmach 2001
rt_tau	consistency	0.04	0.015	+0.0006	18	Ex-Gaussian tau rises with age: West et al. 2002
rt_cv	consistency	0.14	0.035	+0.0012	18	Intra-individual variability: Hultsch et al. 2002
ecc_slope	speed	0.1	0.04	+0.0015	22	Useful field of view narrowing: Ball et al. 1988
peak_speed	control	1.9	0.3	-0.007	20	Peak reach velocity: Ketcham and Stelmach 2001
path_efficiency	control	0.95	0.02	-0.0006	22	Prior, straightness of reach
smoothness	control	-6	0.6	-0.02	20	LDLJ metric: Balasubramanian et al. 2015
choice_rt	decision	0.43	0.05	+0.0025	13	Choice RT age curve: Der and Deary 2006
decision_time	decision	0.13	0.04	+0.001	18	Choice minus simple RT, derived
commission_rate	decision	0.06	0.04	+0.0006	26	Go/no-go errors: Bedard et al. 2002
omission_rate	display	0.02	0.03	+0.0002	–	Quality flag only
d_prime	decision	3.2	0.6	-0.012	22	Signal detection on go/no-go
corsi_span	memory	6.2	1	-0.03	16	Corsi span norms: Kessels et al. 2000, 2008
corsi_total	memory	52	16	-0.45	17	Corsi total score: Kessels et al. 2008
corsi_tap_interval	display	0.55	0.12	+0.004	–	Display only
pursuit_rms_cm	control	4	1	+0.045	16	Prior values, needs calibration
pursuit_lag_ms	speed	120	30	+1.5	18	Visuomotor delay, prior values
pursuit_on_target	display	0.55	0.15	-0.004	–	Display only, redundant with tracking error
pursuit_gain	display	1	0.08	-0.0005	–	Display only

PRIORS

Every norm is provisional. The cited studies set curve shape. None used a headset reach task, so values need refitting on collected sessions (specs/age-model.md v1).

VALIDATION

Ridge model with leave-one-out CV. Report MAE next to the predict-the-mean baseline, and gap-age correlation for regression to the mean.

RELIABILITY

Test-retest on 5 participants. Practice effects held by the fixed practice round.

SCOPE

Functional age for research and demos. No clinical claim or diagnosis.